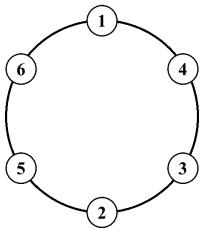
Problem B
Bidirectional Number Ring

Time limit: **1 second**
Memory limit: **1024 megabytes**

Problem Description

A ring is composed of N circles as shown in a diagram. Put natural numbers $1, 2, \cdots, n$ into each circle separately, and the sum of numbers in two adjacent circles should be a prime. Please write a program to find all possible method of filling the number in the circle.



Input Format

The input contains several test cases. Each test case contains one integer N in a line.

Output Format

The output format is shown as sample below. Each test case output all possible methods. Each possible method output two lines, one line represents the numbers in the circles of the ring beginning from 1 in clockwise direction. The other line represents these numbers in anticlockwise direction. Notice that the number of first circle in the ring should always be 1. And the order of numbers must satisfy the above requirements.

Technical Specification

- $2 \leq N \leq 16$

Sample Input 1

```
6
8
```

Sample Output 1

```
Case 1:
1 4 3 2 5 6
1 6 5 2 3 4

Case 2:
1 2 3 8 5 6 7 4
1 2 5 8 3 4 7 6
1 4 7 6 5 8 3 2
1 6 7 4 3 8 5 2
```

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